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Optimization Scheme for Privacy Protection on Cloud Platforms Based on Data Classification

GONG Shuai[1],DONG Xiao Ling[1],YU Dong Bo[1],WANG Hai Lu[1],WEN Zhen Chuang[2],TAO Jun[2]
1. Information and Communication Branch of State Grid Anhui Electric Power Co., Ltd.,Heifei 230041,China
2. School of Cyber Science and Engineering, Southeast University, Nanjing 211189, China

***Abstract*—In addressing how to effectively utilize the powerful features of cloud platforms to provide users with efficient and convenient services while safeguarding the security and privacy of user data, this paper proposes a cloud platform privacy protection scheme based on data classification, namely the PPP-DCP scheme. Through experiments, it is demonstrated that as the user's project set continues to grow and the classification ratio varies, the PPP-DCP scheme outperforms the PrivRank and PrivCheck schemes, offering superior user privacy protection.**

Keywords—Data security; Privacy protection; Privacy issues; Cloud computing technolog

I. INTRODUCTION

With the rapid development of Internet technology, users frequently share their publicly available information such as articles and comments on various online platforms, which are referred to as user-generated public information. The collection and use of these data have raised a series of privacy concerns, involving the protection of user privacy rights and the risks associated with data security [1~3]. Preference privacy protection refers to the capability and control an individual has to protect their personal preferences and interests. It involves the confidentiality and privacy of an individual's preferences, interests, opinions, and other personal characteristics [4~7]. In the digital age, personal preference information becomes a valuable resource for business and marketing activities [8]. For some users, excessive collection and use of preference information may be perceived as an invasion of their privacy[9]. Therefore, safeguarding the privacy of publicly shared information before it is transmitted to online platforms is of utmost importance.

Existing methods for obscuring and protecting the privacy of user-generated public information may result in the loss of the original data's order[10], which can hinder users from receiving high-quality recommendations provided by the platform when using ranking-based personalized recommendations. Addressing these issues, this paper introduces a preference privacy protection solution, PPP-DCP, based on data classification processing. The goal of this solution is to provide ranking-based personalized recommendation services while maximizing the protection of user privacy, and it has been experimentally tested.

II. SYSTEM MODEL AND RELATED DEFINITIONS

A. System Model

Figure 1 illustrates the system model of this solution. The core of this approach involves first categorizing the user's publicly available data, processing the public information based on different categories, and then transmitting the processed information to a third-party recommendation platform to receive high-quality recommendation services.

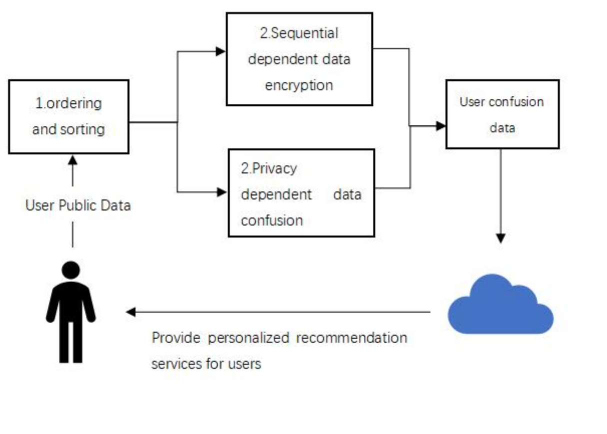

Fig. 1.PPP-DCP System Model.

After the processed data is provided to the third-party recommendation platform, the user can then enjoy personalized recommendation services.

B. Related symbols and definitions

The important symbols used in this article are uniformly explained in Table 1.

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TABLE I. SYMBOL DESCRIPTION

Symbol	Description
C	User disclosure data
r	Classification ratio
P	Preference data set
N	Generic data set
d	Preferred data volume in The original data set
key_table	Key list
D	Plain text field
n	User public data set length
R	Cipher text field
m	Plain text
c	Cipher text
$HGD(M,N,y)$	hypergeometric distribution sampling
α	random number
μ、φ	parameter
N'	Ordinary data set after noise confusion

III. PREFERENCE PRIVACY PROTECTION SOLUTION BASED ON DATA CLASSIFICATION

A. Data Sorting and Classification

User evaluations of an item and their labeling of products can clearly reflect the extent of a user's preference for a particular entity. As shown in Figure 2, the ratings highlighted in red represent the user's preference data, while the remaining ratings are considered ordinary user data.

Film	Score
3 idiots	4.5
The Wandering Earth	1.8
Flipped	3.2
Young-Justice-Bao	2.7

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3 idiots	4.5
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Fig. 2.Sorting and Classification

Algorithm 1 outlines the specific implementation process of sorting and classifying user-generated public data.

Algorithm 1 Sorting and Classification

Input: C, r, n
Output: P, N

```
Initialize p and n as null;
for i in C_1, C_2, ..., C_n do
    Sort i from largest to smallest;
    The amount of preference data d obtained from C × r ;
    if d ≠ 0 then
        The first d preference data in C are put into P;
        The remaining common data in C is stored in N;
    end if
end for
return P, N;
```

B. Sequential Dependent Preference Data Encryption

The sequential dependent encryption and decryption algorithm proposed in this paper utilizes the concept of the hypergeometric distribution. The fundamental idea of the hypergeometric distribution is as follows[11]: within a given ball-drawing model, suppose there are N balls, among which M are black balls, and the remaining $N-M$ balls are white. Each ball is randomly drawn without replacement. In this paper, a random variable X is used to represent the number of black balls obtained in y draws. Therefore, the probability formula (1) for X being equal to x is as follows:

$$P_{HGD}(x;N,M,y)=\frac{\binom{y}{x}\cdot\binom{N-y}{M-x}}{\binom{N}{M}} \tag{1}$$

The sequence-dependent data encryption algorithm is shown in Algorithm 2.First of all, it calculates the value of M、N、d、r、y based on the preference data set P and cipher text field R, the value of M is judged, if $\mathrm{M}=1$, P contains only one data point, then randomly select a number in the range $[\mathrm{r}+1, \mathrm{r}+\mathrm{N}-1]$ and assign it to c, and return it directly; if $\mathrm{M} \neq 1$, then it is judged whether the first column of key_table contains the value of $r+y$, and if it does, the index of that value is found in key_table and the index is saved in $index$. Then set x to the value of the second column of the corresponding index in key_table minus d; If not, use the formula (1) to map the value of $r+y$ to key_table. Then adjust the range of the plaintext field D and the ciphertext field R, and repeat

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multiple rounds of sampling until the value to be encrypted in the plaintext field D is matched. In this case, the value corresponding to the ciphertext field R is the encrypted value.

The sequence-dependent data decryption algorithm is shown in Algorithm 3. The decryption process is similar to the encryption process and will not be described in detail here.

Algorithm 2 Encryption algorithm

Input: key_table, D, R, m
Output: key_table, D, R, c
$M \leftarrow \max(D) - \min(D) + 1; N \leftarrow \max(R) - \min(R) + 1;$
$d \leftarrow \min(D) - 1; r \leftarrow \min(R) - 1;$
$y \leftarrow \lceil N/2 \rceil;$
if $M = 1$ **then**
  return $c \leftarrow random\{r+1, \ldots, r+N-1\};$
end if
if $key\_table(:,1).iscontain(r+y)$ **then**
  $index \leftarrow key_{table(:,1)}.find(r+y);$
  $x \leftarrow key\_table(index, 2) - d;$
else
  $x \leftarrow HGD(M, N, y);$
  $key\_table \leftarrow [key\_table; r+y, d+x];$
end if
if $m \leq d + x$ **then**
  $D \leftarrow \{d+1, \ldots, d+x\};$
  $R \leftarrow \{r+1, \ldots, r+y\};$
else
  $D \leftarrow \{d+x+1, \ldots, d+M\};$
  $R \leftarrow \{r+y+1, \ldots, r+N\};$
end if
return $key\_table, D, R, c;$

Algorithm 3 Decipherment algorithm

Input: key_table, D, R, c
Output: key_table, D, R, m
$M \leftarrow \max(D) - \min(D) + 1; N \leftarrow \max(R) - \min(R) + 1;$
$d \leftarrow \min(D) - 1; r \leftarrow \min(R) - 1;$
$y \leftarrow \lceil N/2 \rceil;$
if $M = 1$ **then**
  return $m \leftarrow \min(D);$
end if
if $key\_table(:,1).iscontain(r+y)$ **then**
  $index \leftarrow key_{table(:,1)}.find(r+y);$
  $x \leftarrow key\_table(index, 2) - d;$
else
  $x \leftarrow HGD(M, N, y);$
  $key\_table \leftarrow [key\_table; r+y, d+x];$
end if
if $c \leq r + y$ **then**
  $D \leftarrow \{d+1, \ldots, d+x\};$
  $R \leftarrow \{r+1, \ldots, r+y\};$
else
  $D \leftarrow \{d+x+1, \ldots, d+M\};$
  $R \leftarrow \{r+y+1, \ldots, r+N\};$
end if
return $key\_table, D, R, m;$

With the privacy protection of user data as a prerequisite, this encryption algorithm ensures the feasibility of database comparison operations. Even if an attacker gains access to all ciphertexts, they will not obtain any useful information beyond the order of the ciphertexts.

C. Privacy-Dependent Confounding of Ordinary Data

Privacy-dependent confounding of ordinary data is carried out using the differential privacy method[12]. The core concept of this method involves introducing a certain degree of noise to the data before processing or publishing it to safeguard individuals' privacy information. The magnitude of the noise can be constrained to meet the requirements of differential privacy. When generating the noise, it is essential to satisfy the inequalities of differential privacy. Hence, the *Laplace* mechanism is chosen to generate noise to achieve privacy protection objectives[13].

Laplace noise constitutes a random value following the *Laplace* distribution. In the realm of differential privacy, *Laplace* noise is a common method for adding noise, and its probability density function is represented as shown in equation (2).

$$f(x \mid \mu, b) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}} \tag{2}$$

Therefore, as per equation (2), the probability density function of the Laplace distribution, the cumulative distribution function (CDF) can be calculated as shown in equation (3).

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$$F(x \mid \mu, b)=\begin{cases}\frac{1}{2}e^{-\frac{\mu-x}{b}}, x<\mu \\ 1-\frac{1}{2}e^{-\frac{x-\mu}{b}}, x\geq\mu\end{cases} \quad (3)$$

Since the range of values for the cumulative distribution function is [0,1], it is necessary to generate random values that follow a uniform distribution within the [0,1] interval before generating *Laplace* noise. This *Laplace* noise, which conforms to the Laplace distribution, can be obtained by solving the inverse of the cumulative distribution function[14]. The inverse cumulative distribution function is represented as shown in equation (4).

$$x=\begin{cases}\varphi\ln(2\alpha)+\mu, \alpha<\frac{1}{2} \\ \mu-b\ln(2(1-\alpha)), \alpha\geq\frac{1}{2}\end{cases} \quad (4)$$

Based on this, setting the parameter μ to 0, results in a new *Laplace* inverse cumulative distribution function, and the formula is obtained as shown in equation (5).

$$\boldsymbol{noise}=\begin{cases}\varphi ln(2\boldsymbol{\alpha})\alpha<\frac{1}{2} \\ -\varphi ln(2(1-\boldsymbol{\alpha}))\alpha\geq\frac{1}{2}\end{cases} \quad (5)$$

The pseudo-code for the specific implementation of noise confounding is presented in Algorithm 4.

Algorithm 4 Noise confusion

Input:N, α, Φ, m

Output:Common user data after noise confusion N'

1：Initialize N' as empty;

2：for i in $N_1, N_2, ..., N_m$ do

3： Assign a random number between 0 and 1 to α;

4： The noise value noise is calculated according to formula (5);

5： $i = i + noise$;

6： Store i in N';

7：end for

8：return N';

Finally, the user data that has undergone confounding is consolidated to create new user public data, thus completing the classification of user public information. Subsequently, the new data is transmitted to the platform for user recommendation services.

IV. EXPERIMENTAL SIMULATION AND RESULT ANALYSIS

In this chapter, the proposed PPP-DCP scheme is experimentally evaluated and compared with the PrivRank framework [15] and the PrivCheck framework [16].

A. Experimental Environment and Evaluation Method

(1)Dataset: The selected dataset for this experiment is the MovieLens dataset, which contains user ratings and movie tag information. It includes four main csv files, with a focus on ratings.csv and movies.csv.

ratings.csv: This file includes user ratings for each movie.

movies.csv: This file contains movie IDs, names, and genres.

(2)Evaluation Methods: The evaluation of the PPP-DCP user preference privacy protection scheme is conducted from two main aspects: privacy protection performance and recommendation quality.

Privacy Protection Performance: In this experiment, privacy protection performance is evaluated by employing inference attacks[17]. Inference attacks are a method of attacking private data, using publicly available information and inference techniques to deduce originally undisclosed sensitive data. This study utilizes the Naive Bayes classification algorithm as the inference attack method.

The process involves selecting 60% of the user data from the dataset as training samples, while the remaining 40% of user data is processed using the PPP-DCP scheme and used as the test dataset. Subsequently, an inference attack is carried out on the test data to assess the classifier's accuracy. In the context of personalized recommendation systems, *precision* is an important evaluation metric that measures the proportion of users with accurately predicted private data in the test dataset[18]. The formula for calculating *precision* as shown in equation (6).

$$precision=\frac{TP+TN}{TP+FP+TN+FN} \quad (6)$$

Where TP represents the number of true positive samples correctly classified as positive, TN represents the number of true positive samples classified as negative, FP represents the number of negative samples incorrectly classified as positive, and FN represents the number of positive samples incorrectly classified as negative. The *precision* values are the average of 10 repeated experiments.

Recommended quality analysis. It refers to the quality analysis of recommendations based on ranking. Selecting

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the POI as the actual scenario enables you to recommend a list of POIs to the user that they are interested in[19].

To assess the recommendation system's performance, Mean Average Precision (MAP) is used to measure the ranking results. A higher MAP value indicates that the recommendation system can more accurately rank items of interest to users, improving the quality of recommendations and user satisfaction[20]. The formula for calculating MAP is as shown in equation (7).

$$MAP = \frac{1}{N}\sum\frac{1}{K}\sum \Pr ecision@k * \mathrm{Re}levant@k \quad (7)$$

Where N is the number of users, k is the length of the recommendation list, *Precision@k* is the accuracy in the top k recommendations, and *Relevant@k* is the number of relevant items in the top k recommendations.

Additionally, precision P and recall R are used to evaluate the recommendation system's quality. P represents the ratio of correctly classified positive samples to the total samples classified as positive by the classifier, while R represents the ratio of correctly classified positive samples to the total number of actual positive samples. The formulas for calculating P and R are as shown in equations (8) and (9).

$$P = \frac{TP}{TP + FP} \quad (8)$$

$$R = \frac{TP}{TP + FN} \quad (9)$$

To provide a comprehensive assessment of both precision P and recall R, the *F1-score* is used. The *F1-score* is the harmonic mean of precision and recall, defined as shown in equation (10).

$$F1 - score = \frac{2 * P * R}{P + R} \quad (10)$$

When the *F1-score* is relatively large, it indicates that the classification model has achieved a good balance between precision P and recall R when considering both factors simultaneously.

B. *Experimental Comparison*

(1) Privacy Protection Performance Analysis: In this section, we evaluate the privacy protection performance of the PPP-DCP scheme, the PrivRank scheme, and the PrivCheck scheme. We employ the Naive Bayes classification algorithm to launch inference attacks on the data that has undergone processing by these schemes. We use *precision* index to quantitatively measure user privacy leakage and assess the privacy protection performance of these three schemes.

TABLE II. USER PRIVACY DISCLOSURE DEGREE

scheme	precision
PrivRank	73.11%
PrivCheck	72.97%
PPP-DCP（r=5%）	68.92%
PPP-DCP（r=10%）	69.48%
PPP-DCP（r=15%）	68.27%
PPP-DCP（r=20%）	69.31%
PPP-DCP（r=25%）	69.13%
PPP-DCP（r=30%）	69.74%

The results in Table 2 show that the accuracy of user data processed by the PPP-DCP scheme is significantly lower when subjected to attacks compared to data processed by the PrivRank and PrivCheck schemes.

(2)Recommendation Quality Analysis: This study primarily focuses on the evaluation of recommendation quality using the metrics of *F1-score* and *MAP*. To assess the performance of the PrivRank, PrivCheck, and PPP-DCP schemes under changes in the user item set *I*, we concentrate on these two metrics. While keeping the values of other parameters consistent, the changes in values will be observed, and the user classification ratio *r* will be used as the cut-off point to conduct a more detailed recommendation quality analysis.

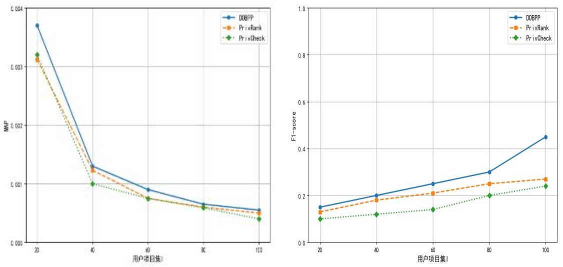

Fig. 3. (a) User classification ratio $r = 10\%$

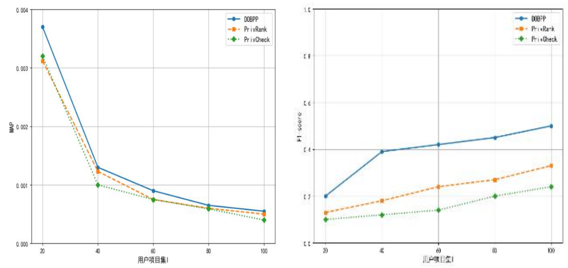

Fig. 3. (b) User classification ratio $r = 15\%$

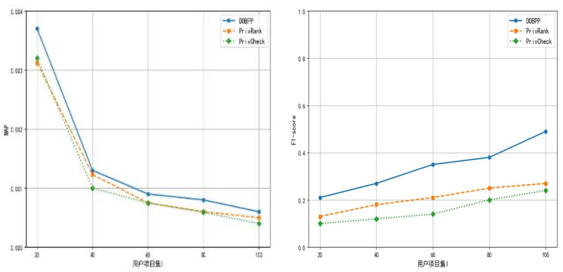

Fig. 3. (c) User classification ratio $r = 20\%$

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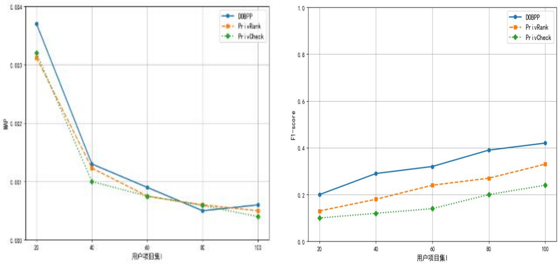

Fig. 3. (d) User classification ratio $r = 25\%$

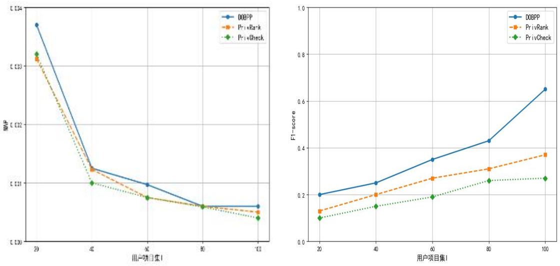

Fig. 3. (e) User classification ratio $r = 30\%$

Figure 3 shows that the PPP-DCP scheme performs well when the user's project set is growing and the classification ratio r is different. Compared with PrivRank and PrivCheck, the data processed by PPP-DCP significantly improves the quality of recommendation service. The values of the PPP-DCP scheme also perform well, much higher than those of the PrivRank scheme and PrivCheck scheme. The user data processed by PPP-DCP can be better applied to recommendation service.

V. CONCLUSION

This paper has proposed a user preference privacy protection scheme PPP-DCP, based on data classification. It has detailed the system workflow and related algorithms of this scheme. By classifying the data generated by users, PPP-DCP ensures that high-quality recommendation services can be provided on the premise of protecting the privacy of users and avoiding inference attacks.

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